

User Submissions Analysis

Practice Questions & Solutions

Q1. List All Distinct Users and Their Stats

Return the user name, total submissions, and total points earned by each user. A list of users with their submission count and total points.

Solution:

```
SELECT
    user_id,
    user_name,
    COUNT(id)           AS total_submission,
    SUM(points)         AS total_points,
    ROUND(AVG(points),2) AS avg_points
FROM resume
GROUP BY 1,2
ORDER BY 3,4 DESC;
```

Q2. Calculate the Daily Average Points for Each User

For each day, calculate the average points earned by each user. A report showing the average points per user for each day.

Solution:

```
SELECT
    submitted_at::DATE,
    user_id,
    user_name,
    ROUND(AVG(points),2) AS avg_points
FROM resume
GROUP BY 1,2,3
ORDER BY 1 ASC;
```

Q3. Find the Top 3 Users with the Most Correct Submissions for Each Day

Identify the top 3 users with the most correct submissions for each day. A list of users and their correct submissions, ranked daily.

Solution:

```
WITH top3 AS
(
```

```

SELECT
    submitted_at::DATE,
    user_id,
    user_name,
    COUNT(id) AS count_submission
FROM resume
WHERE points > 0
GROUP BY 1,2,3
ORDER BY 1 ASC
),
ranks AS
(
SELECT
    *,
    DENSE_RANK() OVER (PARTITION BY submitted_at ORDER BY
count_submission DESC) AS rno
FROM top3
)
SELECT
    *
FROM ranks
WHERE rno <= 3;

```

Q4. Find the Top 5 Users with the Highest Number of Incorrect Submissions

Identify the top 5 users with the highest number of incorrect submissions. A list of users with the count of incorrect submissions.

Solution:

```

SELECT
    user_id,
    user_name,
    COUNT(*) AS incorrect_submissions
FROM resume
WHERE points <= 0
GROUP BY 1,2
ORDER BY 3 DESC
LIMIT 5;

```

Q5. Find the Top 10 Performers for Each Week

Identify the top 10 users with the highest total points earned each week. A report showing the top 10 users ranked by total points per week.

Solution:

```
WITH top10 AS
(
SELECT
    DATE_TRUNC('Week',submitted_at::DATE)::DATE AS firstdateofweek,
    DATE_PART('Week',submitted_at::DATE)        AS week_number,
    user_id,
    user_name,
    SUM(points) AS total_points
FROM resume
GROUP BY 1,2,3,4
),
ranks AS
(
SELECT
    *,
    DENSE_RANK() OVER (PARTITION BY firstdateofweek ORDER BY
total_points DESC) AS rno
FROM top10
)
SELECT
    *
FROM ranks
WHERE rno <= 10;
```